DEPARTMENT OF PSYCHOLOGY AND NEUROSCIENCE

PSYCHOLOGY AND NEUROSCIENCE 50TH ANNUAL GRAHAM GODDARD IN-HOUSE CONFERENCE

Friday, May 15, 2026

Life Sciences Centre
Rooms 4260 and 4332

IN-HOUSE SCHEDULE

9:00 – 9:30		**OPENING REMARKS** Location: **LSC 4260**
	9:00 – 9:10	**Aaron Newman** Opening Remarks
	9:10 – 9:25	**Ray Klein** 50+ Years in Psychology (and Neuroscience) at Dalhousie

9:30 – 10:30 CLICK FOR SESSION A ABSTRACTS		**SESSION A | Psychology, Personality, and People** Session Chair: **Mikaela Ethier-Gagnon** Location: **LSC 4260**
A-1	**9:30 – 9:45**	**Erin Sparks** and **Ella Parsons** Remembering Kailey Simon (2000–2025)
A-2	**9:45 – 9:55**	**Julie Blais** Personalized Persuasion? Testing the Efficacy of Personality-Targeted Political Advertisements
A-3	**9:55 – 10:05**	**Isabella Bossom** Explaining Unique and Common Variance in Financially Focused Self-Concept from Dark Triad and Honesty-Humility Personality Traits Using Commonality Analyses
A-4	**10:05 – 10:15**	**Christopher Park** The Emotional Demands of Civilian Crisis Work: Exploring Burnout, Role Clarity, and Support in Civilian Crisis Intervention Teams
A-5	**10:15 – 10:25**	**Sasha Stephen** Experimental Manipulations to Improve Participant Comprehension of Informed Consent Forms

10:30 – 11:00	**COFFEE BREAK** Location: **LSC 4332** (Lilyan E. White Study Centre) ***Please return to LSC 4260 at 10:55***

11:00 – 12:00 CLICK FOR SESSION B ABSTRACTS	**SESSION B | Attention and Computational Approaches** Session Chair: **Mikaela Ethier-Gagnon** Location: **LSC 4260**

B-1	11:00 – 11:10	**Jeremy Webster** Experimental Investigations of Smartphone Use and Attention
B-2	11:10 – 11:20	**Reyhana Wilson** The Effects of Acute Short-Form Video Viewing on Go/No-Go Performance
B-3	11:20 – 11:30	**Ralph Redden** From Bonding to Focus: Examining the Relationship Between Attachment Styles and Networks of Attention
B-4	11:30 – 11:40	**Nicholas Murray** Assessing Attentional Bias for Cannabis Activated by Trauma Cue Exposure in Cannabis Users with Trauma Histories
B-5	11:40 – 11:50	**Ryan MacArthur** Developing a Deep-Learning Model for Individual Recognition in Canadian Geese: Extending Avian Observational Research Through Computer Vision
B-6	11:50 – 12:00	**Can Sozuer** Quantitative Data-Based-Evaluation of Signal-to-Noise in Assessments Reveals Novel Insights

12:00 – 13:30 CLICK FOR POSTER ABSTRACTS	**LUNCH AND POSTER SESSION** Location: **LSC 4332** (Lilyan E. White Study Centre) ***Please return to LSC 4260 at 13:25***

13:30 – 14:30 CLICK FOR SESSION C ABSTRACTS	**SESSION C | Neuroscience, Vision, and Language** Session Chair: **Blair Irish** Location: **LSC 4260**

C-1	13:30 – 13:45	**Richard E. Brown** The Influence of Ramon y Cajal and Rafael Lorente de Nó on Donald O. Hebb
C-2	13:45 – 13:55	**Andrew Butler** Determining Cortical Roles in Motion Perception via Optogenetic Inactivation of Visual Areas

C-3	**13:55 – 14:05**	**Jeffrey Locke** Seeing More in the Electroretinogram: Integrating Time, Frequency, and Resting-State Analyses Across Mouse Models and Human Retinal Disease
C-4	**14:05 – 14:15**	**Hope Lewis** How Quickly Does Neural Activity Change in Response to Learning Unfamiliar Language Vocabulary?
C-5	**14:15 – 14:25**	**Shuqi Yu** One Skill, Different Journeys? Morphological Awareness and Reading Growth Across Socioeconomic Status

14:30 – 15:00 **COFFEE BREAK**

Location: **LSC 4332** (Lilyan E. White Study Centre)

Please return to LSC 4260 at 14:55

15:00 – 16:00 **SESSION D | Memory, Stress, and Health**

CLICK FOR SESSION D ABSTRACTS

Session Chair: **Blair Irish**

Location: **LSC 4260**

D-1	**15:00 – 15:10**	**Ian Dauphinee** Semantic Similarity Disrupts Order Recall: The Moderating Role of Task Difficulty
D-2	**15:10 – 15:20**	**Mackenzie R. Hartley** The Persistent Effects of Adolescent Chronic Psychological Stress on Sociability in Male and Female Mice
D-3	**15:20 – 15:30**	**Megan Babcock** Expanding on the Two-Hit Model of Depression-Like Behaviour and the Underlying Role of the Oxytocin System
D-4	**15:30 – 15:40**	**Caden Whitfield** The Impact of Income Level on Child Sleep, Parental Engagement, and Perceived Effectiveness of the Better Nights, Better Days eHealth Intervention
D-5	**15:40 – 15:50**	**Mikaela Ethier-Gagnon** The Impact of Menstrual Cycle Phase on Subjective, Behavioural, and Physiological Responses to Electronic Cigarette Ingredients
D-6	**15:50 – 16:00**	**Elena Koning** Psilocybin Treatment as an Adjunct to Cognitive Behavioral Therapy for Eating Disorders: Therapeutic Rationale & Considerations for Protocol Development

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SESSION A TALK ABSTRACTS

A-1 **Erin Sparks** and **Ella Parsons**

Remembering Kailey Simon (2000–2025)

Kailey Simon was from Elsipogtog First Nation in New Brunswick. She was a student in our program, working toward her BA in Psychology with a Certificate in Indigenous Studies. Her involvement at Dal spanned widely: She was a TA for Indigenous Studies, an RA on a project with Math & Stats and Oceanography, and worked with the Office of Sustainability, contributing art that you'll find on signage for the pollinator garden behind our building. Kailey was a persistent student who took intellectual ownership of her work in ways I've rarely seen, and who never shied away from a challenge in pursuit of her academic goals. Her peers in the Indigenous Student Society memorialized Kailey saying, "[s]he brought light, laughter, and care into our community … Kailey was so proud to be L'nu, and her confidence in who she was helped many others feel comfortable in their own skin. Her legacy will live on in everyone who had the privilege of knowing her." I had the privilege of working with Kailey for three years, across contexts. In this talk I'll introduce you to Kailey as I knew her, drawing from my experience watching her navigate my classes and our undergraduate program.

A-2 **Julie Blais,** Scott Pruysers, and Luke Mungall

Personalized Persuasion? Testing the Efficacy of Personality-Targeted Political Advertisements

Before we can evaluate the harm that psychographic microtargeting may entail, we must first empirically examine whether political advertising can be tailored to resonate with people at the level of their personality. In Study 1, we conducted a survey among personality experts (*N* = 86) to evaluate the face validity of real-world targeted ads used for a US midterm election. In Study 2, we conducted a pre-registered between-subjects experimental design with a nationally representative sample of over 2,000 US citizens to assess whether advertisements that match participants' personality traits are more effective (i.e., vote intention, likability, etc.). Study 1 indicated that a majority (63%) of experts in personality psychology were unable to correctly identify the Big Five trait that was targeted in the video. In Study 2, there were no significant aggregate effects of the matching variable, once controlling for participant personality. However, a classification tree analysis did identify a small sub-group of participants where matched ads had higher efficacy. While concerns about psychological voter manipulation are widespread, the results of these studies suggest that such fears are likely overstated. However, even small and subtle effects when scaled to tens of millions of voters may influence election outcomes.

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A-3 **Isabella Bossom,** Sarah Enouy, Julie Blais, Michael Wohl, Scott Pruysers, and Nassim Tabri

Explaining Unique and Common Variance in Financially Focused Self-Concept from Dark Triad and Honesty-Humility Personality Traits Using Commonality Analyses

We investigated the personality correlates of financially focused self-concept, a construct describing individuals who overemphasize financial success as central to their self-definition and self-worth. Although some of the consequences of a financially focused self-concept have been studied, little is known about the personality characteristic of financially focused people. To address this gap, we examined the unique and shared associations between financially focused self-concept and the Dark Triad traits (Machiavellianism, psychopathy, and narcissism) as well as Honesty-Humility at the facet level to provide greater specificity than trait-level analysis. Using cross-sectional data from 1,725 Canadian adults who completed validated self-report measures, we conducted commonality analyses. Results revealed significant associations between financially focused self-concept and Machiavellianism facets (antagonism, planfulness), all psychopathy facets, narcissistic vulnerability, and lower levels of Honesty-Humility (greed-avoidance and modesty). These findings enhance understanding of the personality profiles of individuals who prioritize financial success excessively.

A-4 **Christopher Park,** Chris Giacomantonio, and Lori Wozney

The Emotional Demands of Civilian Crisis Work: Exploring Burnout, Role Clarity, and Support in Civilian Crisis Intervention Teams

Civilian Crisis Intervention Teams (CCITs) are rapidly expanding as alternatives to police-led mental health crisis response across Canada and the United States, yet little is known about responder well-being. Grounded in the Job Demands-Resources (JD-R) framework, this study examined burnout, perceived organizational support (POS), and role clarity during police collaboration among CCIT responders. Participants ($N = 50$) were current or recent civilian crisis responders who completed an anonymous online survey including the Copenhagen Burnout Inventory, the Perceived Organizational Support Scale-10, the Brief Role Clarity in Police Collaboration Scale, and demographic items. Descriptive statistics, reliability analyses, Pearson correlations, and hierarchical multiple regression were conducted. Responders reported moderate-to-high burnout, with the highest levels in personal and work-related domains. POS was significantly and negatively associated with burnout and remained a significant predictor in a model explaining 28.6% of burnout variance, even after accounting for years of experience. In contrast, role clarity did not significantly predict burnout, despite being positively associated with POS. These findings suggest POS may be a central factor in understanding burnout among CCIT responders. As CCIT's continue to expand, strengthening POS may be critical for sustaining CCIT models.

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A-5 **Sasha Stephen,** Sakshi Walia, and Sean P. Mackinnon

Experimental Manipulations to Improve Participant Comprehension of Informed Consent Forms

Though informed consent is a critical part of research ethics, research suggests that comprehension is low. The present study explores methods to improve the comprehension of consent forms via supplementary quizzes and videos. Using an online survey, undergraduates (n = 533) were given a standard consent form. Each participant was randomized to one of 4 conditions: (a) a control condition with standard consent; (b) adding a practice quiz; (c) watching a summary video with a male or (d) female actor. The primary outcome was scores on a multiple-choice quiz of facts related to the consent form. A secondary outcome was whether participants chose to share their data in a repository. Both video conditions were superior to control, with the male video condition having the best comprehension. The quiz condition was not statistically different from control. Though comprehension on most questions was good (~70-80%), understanding of open-access repositories was generally poor (~40%). Despite poor understanding, data sharing in repositories was high (~94%) and there were no differences by condition. Using video summaries following a written consent form can improve comprehension of the consent form's content. However, more work needs to be done to improve participants comprehension of broad consent.

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SESSION B TALK ABSTRACTS

B-1 **Jeremy Webster** and Raymond Klein

Experimental Investigations of Smartphone Use and Attention

A review of research and literature on the interaction between smartphone use and attention. Includes an analysis of present limitations in the field when aiming to assess causality. Review of most commonly practiced experimental designs on the subject: notification, mere presence (brain drain), acute use, prolonged use. Results are interpreted using a contemporary taxonomy of attention (Klein, 2022), and suggest a negative influence on endogenously allocated attention and executive control. Consideration of future directions concludes the review.

B-2 **Reyhana Wilson**, Heather Neyedli, and Cory Munroe

The Effects of Acute Short-Form Video Viewing on Go/No-Go Performance

Short-form videos (SFVs), typically lasting from a few seconds to a few minutes, have become increasingly popular across social media platforms. Although SFV use has been associated with poorer cognition, particularly inhibitory control and attention (Nguyen et al., 2025), experimental evidence testing its acute effects remains limited. This study examined whether brief SFV viewing, relative to a reading control, impairs Go/No-Go task performance, which requires response inhibition and sustained attention. Using a counterbalanced crossover design, participants (N = 36, M_{age} = 18.6, SD = 1.38) completed two 15-minute smartphone-based conditions: SFV viewing and reading. Immediately after each condition, participants completed a Go/No-Go task and self-report questionnaires. Preliminary analyses showed that participants reported an average of 31.8 hours (SD = 15.05) of smartphone use per week, with 44% of this time spent viewing SFVs (M = 14.07, SD = 9.98). The most viewed SFV genres included entertainment, music, people/blogs, comedy and news/politics. Our sample reported substantial SFV viewing, highlighting the popularity of this form of media. The effects of acute SFV viewing on Go/No-Go performance will also be presented.

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B-3 Kyra Battist, Veda Haggarty, Julia Byron, and **Ralph Redden**

From Bonding to Focus: Examining the Relationship Between Attachment Styles and Networks of Attention

Relationship attachment style has been found to influence performance associated with two of the three networks of attention. Attachment avoidance has been shown to influence the executive network of attention, wherein individuals higher in attachment avoidance show reduced interference in the standard Eriksen Flanker task. Attachment anxiety has been shown to influence the executive network as well, in addition to the orienting network of attention, wherein individuals higher in attachment anxiety show larger endogenous cueing effects. The current study was designed to replicate these prior findings, and to examine the heretofore untested relationship between attachment styles and the alerting network of attention, all in a single task. AttentionTrip is an engaging, gamified version of the Attention Network Task, played on an iPad tablet. Attachment styles were assessed using the Experience in Close Relationships-Revised Questionnaire. Robust network scores were observed for each network of attention. Attachment styles were associated with inverse relationships with overall RT. Moreover, orienting and executive control interacted differently with both attachment styles, whereas neither attachment style reliably influenced the alerting network. These findings support the hypothesis that the two forms of insecure attachment are associated with different underlying attentional functioning.

B-4 **Nicholas Murray,** Sarah DeGrace, Melissa Burke, Isabella Bossom, Mikaela Ethier-Gagnon, Raymond Klein, and Sherry Stewart

Assessing Attentional Bias for Cannabis Activated by Trauma Cue Exposure in Cannabis Users with Trauma Histories

Posttraumatic stress disorder (PTSD) and cannabis use disorder (CUD) co-occur at rates five times higher than chance. An integrated self-medication/attentional bias (iSM-AB) model suggests that trauma reminders may shift attention toward cannabis cues in cannabis users with trauma histories, increasing cravings and intentions to use. This project tests the iSM-AB model using a personalized cue reactivity paradigm and an Addiction Stroop Task. To date, 17 (aiming to continue recruitment for a total of 50) participants completed both sessions. In Session 1, participants completed questionnaires and a semi-structured interview to generate personalized trauma and neutral scripts. In Session 2, participants were exposed to trauma and neutral cues, each followed by an Addiction Stroop Task with cannabis-related and neutral words. Preliminary linear mixed model results indicate that among individuals with higher PTSD symptoms, trauma cue exposure (but not neutral exposure) increases cannabis-related attentional bias ($\beta = 0.43$, $p = .04$). Post-cue negative affect was also positively associated with attentional bias, but only in the trauma condition ($r = 0.46$, $p = .03$). These preliminary findings support the iSM-AB model and suggest attentional bias modification training may be best targeted to individuals with more severe PTSD symptoms when preceded by trauma cue exposure.

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B-5 Ryan MacArthur

Developing a Deep-Learning Model for Individual Recognition in Canadian Geese: Extending Avian Observational Research Through Computer Vision

Individual recognition is a foundational cognitive pillar that supports complex social architectures across the animal kingdom, enabling behaviours such as long-term pair bonding, reciprocal altruism, and hierarchy maintenance. Studying these behaviours requires both establishing cognitive capacity and developing non-invasive tools to observe undisturbed behaviour. Avian ethology has emphasized auditory recognition, yet mounting evidence shows visual signals also carry identity information. Work on the Greylag Goose (*Anser anser*) has demonstrated bill-and face-level individuality with high algorithmic accuracy. The Canadian Goose (*Branta canadensis*) presents as a similar candidate and remains unexplored. Here we report a computational study comparing two deep-learning architectures trained and tested on identical data, as a step toward non-invasive individual identification. Individually-labelled images were assembled from a reference dataset and from video-derived exports produced by a purpose-built annotation tool. We tuned a ResNet-50 classifier (closed-set assignment) and a MegaDescriptor-L-384 retrieval model (gallery matching). Both reached >90% accuracy on novel test images. Saliency maps emphasized the cheek-patch boundary, bill, and facial regions as candidate identity cues. Together, the results support a passive, camera-based alternative for avian observational research.

B-6 Can Sozuer

Quantitative Data-Based-Evaluation of Signal-to-Noise in Assessments Reveals Novel Insights

Online learning platforms like Brightspace, Remark, or Crowdmark can display evaluative metrics for questions graded on their platforms. These metrics estimate how well a question assesses student learning by comparing how well performance on that question predicts performance on other questions (signal). However, platform-provided metrics may not evaluate well assessments with a wide variety of question types, as this variation reduces predictiveness. They also cannot incorporate question information supplied externally by a teaching user to account for between question variation. Here, I present a new metric for evaluating how well a question assesses learning, with the added ability to incorporate externally provided question information. This metric is thought to evaluate how well a question assesses learning, especially on high-variation assessments, more accurately than readily available alternatives. I then apply this metric to datasets to extract insights into which assessment design elements increase how well a question assesses student learning (signal).

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SESSION C TALK ABSTRACTS

C-1 **Richard E. Brown**

The Influence of Ramon y Cajal and Rafael Lorente de Nó on Donald O. Hebb

Donald O. Hebb obtained his BA in English and Philosophy from Dalhousie University in 1925 and his MA in Psychology from McGill in 1932. He completed his PhD with Karl Lashley at Harvard in 1936. Between 1932 and 1934 he wrote three unpublished papers on the neural basis of behaviour. After his PhD he conducted neuropsychological tests on the patients of Dr. Wilder Penfield at the Montreal Neurological Institute (1937-39). From 1939 to 1942 he was a professor at Queen's University in Ontario and then he studied emotionality in chimpanzees at the Yerkes Primate Centre in Florida (1942-1947). It was here in 1944 that he began to revise his old theory of the neural basis of behaviour and wrote five sets of unpublished notes for his influential book, The Organization of Behavior. This book introduced the concepts of the "Hebb synapse" and learning rule, cell assemblies, and phase sequences to account for the neural events underlying behaviour. The crucial element in Hebb's theory was the reverberatory circuit of Lorente de Nó, based on Cajal's concept of closed circuits. This presentation examines the unpublished notes and papers written by Hebb as he developed his neuropsychological theory.

C-2 **Andrew Butler,** Nicole Michaud, Lan Dang, and Nathan Crowder

Determining Cortical Roles in Motion Perception via Optogenetic Inactivation of Visual Areas

Motion perception is an integrative process that combines information about the direction, size, and velocity of a stimulus. Our aim is to examine the cortical networks underlying speed discrimination, which is the ability to perceive differences in velocity. The optogenetic regulation of an inhibitory interneuron circuit in the mouse cortex allows for reversible inactivation of various cortical areas to determine which areas are involved with speed discrimination. By combining this optogenetic technique with a 2-alternative forced-choice task where the mouse must select the faster of two random-dot kinematograms, we are able to assess if the targeted region is involved in speed discrimination. Pilot data has already been collected showing that optogenetic inactivation of the primary visual cortex (V1) can impact motion perception. From here we plan to conduct a more complete experiment to fully assess the role of V1 and higher visual areas in speed discrimination. This research provides foundational information about the contributions of individual cortical regions to visual perception, but also provides a methodological framework for future research to assess the relevance of individual cortical areas to any behaviourally measurable aspect of perception or cognition due to canonical circuitry.

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C-3 **Jeffrey Locke,** Patrice Côté, and Aaron Newman

Seeing More in the Electroretinogram: Integrating Time, Frequency, and Resting-State Analyses Across Mouse Models and Human Retinal Disease

Interpretation of the electroretinogram (ERG) has traditionally focused on time-domain waveform features which may oversimplify complex retinal signaling. Frequency-and time-frequency-domain analyses offer complementary perspectives but remain inconsistently applied due to methodological variability. Resting-state ERG has also emerged as a potential probe for intrinsic retinal activity, though its relationship to evoked responses remains poorly defined. Full-field ERGs were recorded across stimulus-evoked and resting-state conditions in mouse models representing normal retina (n=23), inner-retinal dysfunction (n=10), and global-retinal degeneration (n=11). Signals were analyzed using time-domain metrics, power spectral density, and time-frequency transforms. Time-frequency analysis employed Morlet wavelets parameterized using a full-width-at-half-maximum (FWHM) framework to explicitly control temporal-spectral trade-offs. An identical analytical pipeline was applied to human clinical ERG and resting-state data (n=69). Progressive retinal dysfunction produced graded reductions in waveform amplitude, spectral power, and time-frequency energy, with broad, permutation-corrected spectral and temporal clusters distinguishing phenotype. Resting-state ERG demonstrated consistent alterations in aperiodic spectral slope and high-frequency structure, persisting even when evoked responses approached the noise floor. Comparable patterns were observed across species. Integrating time-domain, frequency-domain, time-frequency, and resting-state analyses provides a more sensitive and comprehensive assessment of retinal function than any signal method alone.

C-4 **Hope Lewis,** Cate Brandt, and Aaron Newman

How Quickly Does Neural Activity Change in Response to Learning Unfamiliar Language Vocabulary?

Semantic processing is associated with the N400 event-related potential (ERP), with larger N400 amplitudes appearing when information mismatches versus when it matches expectation. This differentiation is called the N400 effect, which does not appear with unfamiliar language (UL) vocabulary before training, but does appear by 10 days after training begins. By recording EEG throughout the training process, the current study determined how much exposure to UL vocabulary a new learner needs to show an N400 effect in response to said UL vocabulary. Over two sessions, Spanish vocabulary was taught to UL learners by pairing words with images representing their meaning. A semantic mismatch task paradigm was used to assess N400 ERPs in Spanish and in English before and after each block of training in each session. We found that after 3-4 exposures to UL words in the first session, an N400 effect emerged, with this remaining during the second session. Mean amplitude was smaller in the Spanish vocabulary compared to English, reflective of the difference in proficiency. Accuracy of responses also improved over time, reaching ceiling by the end of day 1. Findings provide useful insight into the earliest neural markers of acquiring new semantic information with UL vocabulary.

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C-5 **Shuqi Yu** and Hélène Deacon

One Skill, Different Journeys? Morphological Awareness and Reading Growth Across Socioeconomic Status

Reading comprehension supports learning across academic subjects. A robust predictor is morphological awareness (MA), the ability to recognize and manipulate meaningful word units. Although higher-SES children show faster reading growth especially in Grade 1, it remains unclear how SES shapes the long-term impact of Grade 1 MA on reading growth. Therefore, this study examined whether MA contributes differently across socioeconomic contexts from Grade 1 to Grade 4. Participants were 313 English-speaking children from 18 primary schools in Eastern North America. In Grade 1, children completed MA, reading comprehension, and control measures (word reading, vocabulary, phonological awareness, nonverbal ability, working memory). Parents reported family demographics, including SES. In Grade 4, standardized reading comprehension was assessed. Grade 1 MA uniquely predicted Grade 4 reading comprehension ($\beta = 0.51$, $p < .001$). SES was also positively associated with later reading comprehension ($\beta = 0.10$, $p = .01$). A significant MA × SES interaction indicated that the association between MA and reading comprehension growth was stronger for children from lower-SES backgrounds ($\beta = -0.11$, $p = .02$). These findings suggest that MA is particularly important for reading development among lower-SES children, highlighting its potential as a target for early instruction.

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SESSION D TALK ABSTRACTS

D-1 **Ian Dauphinee,** Dominic Guitard, Sho Ishiguro, and Jean Saint-Aubin

Semantic Similarity Disrupts Order Recall: The Moderating Role of Task Difficulty

Similarity affects memory across many stimulus domains, typically impairing short-term order recall, with one important exception: semantically similar words (e.g., apple, orange, banana). In fact, the challenging insensitivity of short-term order recall performance to semantic similarity is has become a benchmark effect for evaluating memory models. However, contrary to the literature, three key studies reported a detrimental effect of semantic similarity on short-term order recall. Here, in five large-scale experiments, we revisited these three exceptions. Despite implementing more rigorous methodological controls, we reproduced the detrimental effect of semantic similarity across all experiments. We then conducted a review and meta-analysis, which revealed a small but reliable detrimental effect across the literature. To account for previous inconsistencies, we identified a potential moderating variable: task difficulty. Across four additional experiments, we manipulated task difficulty via presentation speed and list length. As predicted, under more difficult task conditions semantic similarity impeded order recall, and critically, when the task was easier, its detrimental effect on order recall vanished. These findings show that semantic similarity is not a special case for short-term order memory and that its effect depends in part on task difficulty, helping reconcile previously conflicting results for models of serial order memory.

D-2 **Mackenzie R. Hartley** and Tamara B. Franklin

The Persistent Effects of Adolescent Chronic Psychological Stress on Sociability in Male and Female Mice

Adolescents are more sensitive to stress than adults and experiencing stress during this developmental period can have lasting effects on behaviour, including increased depression-like and anxiety-like behaviours. Our research aims to determine the effects of chronic adolescent stress on adult sociability, and the neural mechanisms that may underlie these effects. Our behavioural research has found that adolescent Chronic Predator Stress (aCPS) exposure leads to an increase in same-sex peer investigation and altered social dominance behaviours in adult mice. Gene expression in the brains of these animals was analysed using rt-qPCR analyses, with an initial focus on the dorsal and ventral hippocampus, brain regions that are still developing during adolescence, are responsive to stress, and have been shown to be critical for social function. We quantified Drd2 and Oxtr (genes that encode for dopamine and oxytocin receptors), and histone deacetylase 2 (Hdac2), all of which represent systems that have been associated with both social behaviours and stress. aCPS had no effect on hippocampal expression of these genes in male or female mice. Future work includes a brain-wide c-Fos mapping study to gain insight into the brain areas likely to contribute to the altered peer investigation behaviours observed in our behavioural study.

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D-3 **Megan Babcock,** Kate Zucconi, Janet Menard, and Tamara Franklin

Expanding on the Two-Hit Model of Depression-Like Behaviour and the Underlying Role of the Oxytocin System

The two-hit hypothesis proposes that early-life stress increases vulnerability to later stressors. While this hypothesis was originally developed in schizophrenia research, recent work has applied it to depression as well. My MSc research tested whether early-life adversity (ELA) heightens sensitivity to subthreshold stress. Rats exposed to both neonatal limited bedding and nesting stress and a subthreshold version of chronic mild stress in adolescence displayed greater depression-like behaviour and aggression in adulthood, whereas neither stressor produced effects alone. LBN exposure was also associated with reduced oxytocin (OT)-immunoreactive cells in the paraventricular nucleus, a key OT-producing region that helps regulate stress responses and social behaviour. These findings suggest that ELA renders individuals more vulnerable to depression-like behaviour following a later stressor, even when subthreshold, and implicate altered OT signaling as a potential underlying mechanism. Building on this work, my proposed PhD research will test the hypothesis that ELA disrupts OT system function, and that these disruptions have lasting consequences for stress sensitivity across development. Specifically, I will investigate whether the timing and type of stress exposure differentially alter OT production, release, and receptor signaling, and whether these changes drive depression-like and social behavioural outcomes.

D-4 **Caden Whitfield**, Mya Dockrill, Penny Corkum, Debbie Emberly, Christine Cassidy, Alissa Pencer, and Guido Simonelli

The Impact of Income Level on Child Sleep, Parental Engagement, and Perceived Effectiveness of the Better Nights, Better Days eHealth Intervention

Behavioural sleep disturbances (e.g., insomnia) are common in school-aged children and negatively impact health and functioning. *Better Nights, Better Days (BNBD)* is an evidence-based eHealth intervention proven to improve sleep in children. Given that most intervention research mainly includes participants with middle to high income, we do not know how income level impacts children's sleep and treatment for sleep problems. A secondary analysis of four previous *BNBD* studies was performed to examine differences between families with low income (n = 145) and mid-high income (n = 820) regarding sleep problem severity, engagement with the intervention, and perceived effectiveness. Data from sleep health questionnaires demonstrated that low-income children had more severe sleep problems. User analytics (i.e., number of sessions completed) from the *BNBD* intervention showed very similar rates of engagement between income-level groups. Differences between pre-and post-intervention questionnaires for participants who completed a "clinical dose" of the program showed that the low-income participants reported lower effectiveness than the mid-high-income participants. This study supports that sleep health disparities are present in Canadian children and demonstrates the promise of *BNBD* as a usable intervention for low-income families, while emphasizing the need to further investigate the intervention's effectiveness for low-income families.

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D-5 **Mikaela Ethier-Gagnon,** Makayla MacLean, Robin Perry, Cory Munroe, Tara Perrot, Sherry Stewart, Kim Good, and Sean Barrett

The Impact of Menstrual Cycle Phase on Subjective, Behavioural, and Physiological Responses to Electronic Cigarette Ingredients

Electronic nicotine delivery systems (ENDS) use has increased in recent years, partly due to appealing flavours such as menthol. Sensory factors appear particularly important for nicotine reinforcement in females, and ovarian hormones may modulate these effects. However, the role of the menstrual cycle in menthol-and nicotine-related reinforcement remains largely unexamined. Naturally cycling female ENDS users (N = 34) completed experimental sessions during two menstrual phases (follicular and luteal phase) in a randomized mixed within-subjects (menstrual phase) and between-group (menthol vs. menthol-free) design. Immediately before and after a standardized puffing procedure, subjective measures (craving, withdrawal, subjective state, inhaler effects) and heart rate variability (HRV) were assessed. Salivary estradiol and progesterone were collected to validate menstrual phase and for exploratory analyses. Menthol enhanced sensory and reward-related subjective effects and was associated with greater reductions in relief craving relative to the menthol-free condition. In the menthol-free condition, participants in the follicular phase reported greater withdrawal and subjective ‘high’ relative to the luteal phase. Exploratory analyses indicated that higher progesterone levels were associated with reduced withdrawal in the menthol-free condition only. Results support policy efforts regulating menthol in ENDS products and highlight the need for targeted cessation strategies for female users.

D-6 **Elena Koning,** Susan Gamberg, and Aaron Keshen

Psilocybin Treatment as an Adjunct to Cognitive Behavioral Therapy for Eating Disorders: Therapeutic Rationale & Considerations for Protocol Development

Eating disorders (EDs) remain challenging to treat, with high dropout and low remission rates in cognitive-behavioral therapy for EDs (CBT-ED). Psilocybin treatment (PT) demonstrates therapeutic potential to enhance CBT-ED by exerting several neurobiological (e.g., neuroplasticity, cognitive flexibility), psychological (e.g., antidepressant), and experiential (e.g., illness insights) effects. Specifically, the effects of PT are hypothesized to increase psychotherapeutic engagement/learning, reduce dropout, and improve clinical outcomes. This perspective article (recently published in Behavioral Sciences) provides the first consolidation of neurobiological and transdiagnostic clinical evidence for PT/CBT-ED, proposes considerations for a concurrent intervention protocol, and presents clinical research considerations to empirically test its feasibility, safety, and efficacy. This line of inquiry is expected to advance the development of approaches that improve ED treatment outcomes. More broadly, it advances the study of psychedelics as tools to enhance evidence-based psychotherapy models for difficult-to-treat neuropsychiatric conditions.

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POSTER ABSTRACTS

1 **NESC/PSYO 3370: Sarah MacMillan, Ewan MacKinnon,** Avery Nadalini, Emma Morais, Ella Power, Ozan Ozmen, Laura Kuyper, Brian Corbett, Elijah Akpan, Utkarsh Pandey, Sofia Westfall, Khushbu Aumeer, Max Vaillancourt, Luke Janze, Sarah Taylar, and Kevin Duffy

Synapsin Immunolabeling in the LGN Remains Unchanged Following Monocular Retinal Inactivation

Conditions that disrupt visual experience early in life can alter the structure and function of neurons, producing a vision impairment called amblyopia. Success recovery with conventional occlusion treatment is plagued by poor compliance, and is effective only when administered very in early childhood during the critical period. Previously, we demonstrated rapid recovery from the effects of monocular deprivation following temporary retinal inactivation of the dominant eye with intraocular application of tetrodotoxin. Brief retinal inactivation produced rapid and significant structural and functional recovery at ages beyond what is observed with mainstay occlusion therapy. Verifying the safety of retinal inactivation is of paramount importance in considering it as viable means of correcting the effects of early abnormal vision. In this study we show that immunolabeling for a common marker of synaptic integrity, synapsin, is preserved after retinal inactivation indicating that synapses serving the silenced eye are not lost following temporary loss of retinal activity.

2 **Aarushi A** and Javeria Hashmi

Resilience Profiles in Chronic Pain: The Role of Trauma Exposure & Symptoms

Chronic pain is a prevalent and complex condition that causes physical, psychological and functional difficulties. Such populations usually have had exposure to traumatic events, which have been linked to worse psychological & pain-related outcomes. Individual responses to such traumatic exposure, however, differ greatly: some may develop post-traumatic stress symptoms (PTSS), but others may maintain relatively adaptive daily functioning. These differences suggest that psychological and clinical pain outcomes, specifically after trauma exposure, may be significantly influenced by resilience. The present study aims to examine resilience profiles in individuals with chronic pain, with a particular focus on how trauma exposure and PTSS relate to adaptive traits and pain outcomes. Using pre-existing data from a chronic pain study, we looked at whether those with trauma exposure who report fewer post-traumatic stress symptoms exhibit higher levels of psychological resilience resources, more adaptive acute and clinical pain profiles as well as better emotional burden management. A one-way ANOVA was used to evaluate group differences amongst the 16 variables, with a non-trauma-exposed group included for comparison. Findings revealed that while the resilient and vulnerable groups differed in their psychological resilience resources and emotional burden, there were no significant differences in the clinical pain outcomes. These results suggest that resilience and vulnerability may be more strongly reflected in psychological and emotional domains rather than in pain intensity itself. Hence, this study aims to advocate for more individualized trauma & resilience-informed approaches into chronic pain assessment and management, supporting more individualized and comprehensive care strategies.

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3 **Laura Adams,** Anika Cloutier, Daphna Motro, and Maria Adams

Leaders' Depression Disclosure: Followers' Perceptions and Leadership Expectations

Organizations are increasingly encouraging leaders to foster a positive mental health climate, including by disclosing personal challenges. However, how followers perceive leaders following such disclosure remains unclear. This study examined how the degree of leader's depression disclosure influences follower perceptions and leadership expectations. Drawing on stigma theory and implicit leadership theories (Goffman, 1963; Lord et al., 2020), we predicted that greater disclosure would reduce perceived leader prototypicality, which would in turn lower expectations of effective leadership behaviours. Based on role congruity theory (Eagly & Karau, 2002), we further predicted stronger negative effects for male leaders, given the incongruence between depression and male gender norms. 403 Prolific participants were randomly assigned to a 4 (disclosure degree: control, low, moderate, high) × 2 (leader gender: man, woman) video vignette design. Participants rated leader prototypicality and the expected servant and laissez-faire leadership behaviours. Results indicated that the disclosure amount did not affect prototypicality, nor did prototypicality mediate leadership expectations; these relationships were not moderated by leader gender. However, exploratory analyses showed that authenticity ratings varied by condition, with moderate disclosure rated higher than low, and authenticity mediated the link between disclosure and leadership expectations. Findings contribute to both occupational health and leadership literatures.

4 **Victoria Adams,** Tamara Franklin, Sarah Hughes, Mackenzie Hartley, and Megan Babcock

Comparing the Warm Spot Test and Tube Test as Behavioural Measures of Social Hierarchy in C57Bl/6N Mice

Social hierarchies are fundamental to communal species, shaping social behaviour, stress responses, and cognition. In rodent models, reliable and ecologically valid measures of dominance are essential for accurately linking social rank to biological processes. The widely used tube test establishes hierarchy based on winning/losing over repeated trials across 7–10 days, which may introduce learning effects and confound results. The warm spot test is a newer, less-studied approach that suggests that dominant mice will spend the most time in the warm spot compared to subordinate mice. However, knowledge gaps remain regarding its validity, reliability, and ability to distinguish intermediate (Rank 2) from subordinate (Rank 3) individuals. This pilot study directly compared the warm spot and tube tests in adult male and female mice housed in trios. Mice were assessed using the warm spot test before and after a 7-day tube test protocol. Males performed similarly on both warm spot trials, indicating stability despite tube test exposure, whereas females did not.
Notably, males identified as rank 2 based on number of wins in the tube test spent more time in the warm spot than Rank 1 or Rank 3 mice, suggesting distinct behavioural strategies in this group. Overall, our findings suggest that the warm spot test is more consistent in males than females, and that time in the warm spot does not measure the same dominance trait as number of wins in the tube test.

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5 **Mohammed Ali Ahmed** and Richard E. Brown

Effect of Number of Training Trials on Olfactory Learning and Memory in a Pavlovian Conditioned Odour Preference Task

Olfactory learning is a robust and ethologically relevant model for investigating cognitive function in rodents. However, defining the boundary conditions of these tasks, particularly the minimum number of training trials required to generate a stable memory trace, is essential for developing sensitive behavioural assessments. This study examined these conditions in the Pavlovian Conditioned Odour Preference Task using the Nrxn1+/-mouse model of Autism Spectrum Disorder. Male and female Nrxn1+/-mice and wild-type littermates underwent a one-day training protocol consisting of either 16 or 8 training trials. Memory strength was assessed in 24-hour and 14-day memory tests, followed by extinction trials and a spontaneous recovery test. Results indicated that 8 trials were insufficient to produce an odour preference in either genotype. Unexpectedly, wild-type mice also failed to show a preference in the 16 trial condition, whereas Nrxn1+/-mice demonstrated both short-and long-term memory. However, this memory was extinguished within two unreinforced trials after 14 days and did not show spontaneous recovery. These findings are compared with the results of previous studies in our lab and define memory formation and extinction parameters in the one-day Conditioned Odour Preference Task as an efficient assay for memory strength in mice.

6 **Madison Baggs**

The Impact of High Cognitive Demand on the Directed Forgetting of Negative Words

This experiment used an item-method paradigm to examine whether high versus low cognitive demand during encoding influences directed forgetting of negative and neutral words. Using rapid serial visual presentation, each study trial began with 8–15 random letter strings, followed by a neutral or negative study word and then by an instruction to remember or forget. This was followed by seven additional letter strings, one of which was a probe consisting of all Xs or all Os. In the high cognitive demand condition, participants identified the probe by pressing X or O on the keyboard; in the low cognitive demand condition, participants were not required to identify the probe and, instead, pressed the space bar at the end of each study trial. After all study trials were presented, a yes-no recognition task assessed the directed forgetting effect for negative and neutral study words. Directed forgetting was defined as the difference between to-be-remembered word recognition and to-be-forgotten word recognition. A two-way ANOVA showed a main effect of valence, with a smaller directed forgetting effect for negative words compared to neutral words; a main effect of cognitive demand, with a smaller directed forgetting effect when participants encoded words under high demand compared to when they encoded words under low demand; and, no significant interaction. These results suggest that the influence of negative valence on directed forgetting occurs early in encoding and is not affected by subsequent demands on limited-capacity resources.

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7 **Jiah Bhutani,** Cameron Calder, and Javeria Ali Hashmi

The Role of Trauma History in Chronic Pain: Distinguishing the Effects of Type, Timing, and Cumulative Exposure

Trauma exposure is highly prevalent in chronic pain and may contribute to pain-related symptom burden. However, trauma is often treated as a single construct, limiting understanding of how trauma type, developmental timing, and cumulative exposure differentially relate to pain and psychological outcomes. This study examined these trauma dimensions in adults with chronic pain. Participants were 149 adults with fibromyalgia, chronic back pain, or both. Lifetime trauma was assessed using the Brief Trauma Questionnaire, and developmental trauma was assessed using the Childhood Trauma Questionnaire and the Adverse Childhood Experiences questionnaire. Interpersonal trauma was based on endorsement of relevant items. Outcomes included pain threshold, pain tolerance, pain severity, pain interference, pain catastrophizing, depressive symptoms, state and trait anxiety, post-traumatic stress symptoms, and dissociative symptoms. Trauma exposure was common, with sexual violence, serious accidents, and witnessing trauma most frequently reported. Women reported higher rates of sexual violence, whereas men reported higher rates of combat exposure. Approximately 60% reported at least one interpersonal trauma, which was associated with greater affective and post-traumatic stress symptom burden. Lifetime trauma was associated with higher pain severity and interference, whereas developmental trauma was associated with lower pain tolerance. Findings support a multidimensional, trauma-informed approach to chronic pain assessment and management.

8 **Karina Bouter,** Sophia Giazizidou, Catherine Mimeau, and Hélène Deacon

Do Students with Dyslexia Learn the Spellings and Meanings of Words Through Reading?

Fluent and accurate reading is a critical skill in navigating society. Typically developing children learn the spelling and meaning of words through independent reading, called orthographic and semantic learning respectively. We investigate whether these skills are impaired in students with dyslexia – children with poor word decoding and fluency. We tested orthographic and semantic learning in 139 Grade 3 students. We identified 21 students with dyslexia (scored below one standard deviation on standardized measures of word reading) and matched 21 controls by age and nonverbal intelligence. Students completed a learning task where they read stories about new inventions (e.g., a fish tank cleaner called a *veap*). Students demonstrated orthographic learning by choosing the invention spelling from four nonwords and semantic learning by choosing the image depicting its function from four pictures. We found that controls outperformed those with dyslexia on orthographic learning ($p = .013$, $d = .8$) but not semantic learning ($p = .277$, $d = .34$). Harnessing strengths in semantic learning may help students with dyslexia form orthographic representations of new words, increasing their reading fluency and accuracy.

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9 **Heather Carr** and Simon Gadbois

Assessment of Biomedical Alert Dog Candidate Learning and Training Methodology

Biomedical alert dogs are utilized increasingly in treatment and diagnosis, resulting in an increase in related studies in the recent literature regarding their potential. This growth has led to ever-increasing demand for trained animals. Training protocols are not able to keep up with the need, leading to long wait times for service dogs. Modifications of training procedures to best accommodate dogs' learning mechanisms could streamline this process. The importance of investigating the best way to train a given dog is underscored by the extensive time and economic investment in training a scent-detection dog. In this study, I will examine scent-detection canine performance across two different scent-detection training methodologies. My questions will be whether the matrix technique pioneered in the Wildlife Ethology and Canine Olfaction Lab (W.E.C.O.) offers advantages over the funnel technique for the dog's overall performance, specifically accuracy and precision (consistency in the type of error), as well as if it benefits the overall consistency of their performance.

10 **Ava Cole,** Isaiah Seale, and Richard Brown

Sex Differences in Olfactory Learning and Memory Across Nrxn1 Genotypes in Mice

Autism spectrum disorder (ASD) is a neurodevelopmental disorder affecting ~1% of the population. Neurexins, including Nrxn1, are strongly linked to ASD and have been associated with learning deficits. This study examined sex differences in four Nrxn1 genotypes in mouse models using a Pavlovian conditioned odour preference paradigm, with memory tested 24 hours, 7 days, and 30 days after training. Learning and memory were evaluated through digging behaviour in the rewarded (CS+) and non-rewarded (CS-) odour pots, as well as difference and preference scores. Sugar consumption was also measured during acquisition. Males displayed significantly more digging behaviour than females during training and early memory tests, although preference scores did not differ. No significant genotype effects or sex by genotype interactions were observed. These findings suggest that Nrxn1 related learning deficits may not be detectable using simple Pavlovian conditioning procedures. Observed sex differences indicate that performance strategies may vary while overall memory abilities remain comparable between males and females. High preference scores further indicate that this one-day Pavlovian conditioned odour preference training procedure is effective, and mice can acquire, store and consolidate olfactory memories 24 hours, 7 days, and 30 days after learning the association.

11 **Ella Mae Cueto,** Gia Samuel, Paige Parsons, Ian Dauphinee, and Raymond Klein

The Exploration of Sensitivity and Response Bias in the Vigilance Decrement

Vigilance, or sustained attention over time is essential in numerous fields, and a decline in vigilance during a repetitive, cognitively taxing task is often observed. The exact cause of this decrement is not completely clear. However, in current literature, sensitivity and criterion are two theories that could help explain this decline in hit rate over time. We investigated whether vigilance decrement was due to a sensitivity decline, a conservative criterion shift, or combination of the two. Nine participants were told to imagine themselves in the role of a radar operator scanning spy planes as their targets and passenger planes as their non-targets and asked to perform a repetitive, cognitively demanding task over the course of 35-40 minutes. Participants detected these targets while rating their confidence level. In contrast to our hypothesis, results indicated that participants did not adopt a criterion shift and exhibited an increase in sensitivity. We conclude that because our vigilance task was too cognitively demanding, they did not experience a vigilance decrement.

12 **Rachel Dean,** Kevin LeBlanc, Anne Lacroix, Brett Feltmate, and Heather Neyedli

Planning for the Unknown: Effects of Target Uncertainty on Grasping

Motor planning for grasping known objects is well understood, but how we plan grasping movements under uncertainty is not. Competing theories propose an averaging strategy, where multiple possible actions are combined, or an optimization strategy, where the safest or most demanding movement is prioritized (maximal-safety approach). Research supports the maximal-safety approach, showing larger hand openings when a larger object is present, regardless of the target. However, those findings may be influenced by differences in object size and weight. To address this, we used two identical rectangular objects and varied their orientation to change the required grip size while keeping the physical properties the same. Participants (n = 25) completed two conditions, a Go-Before-You-Know (GBYK) condition, where the target was revealed after movement, and Know-Before-You-Go (KBYG) condition, where the target was specified in advance. Peak grip aperture (PGA) was defined as the maximum distance between the thumb and index finger during the reach. Under uncertainty (GBYK), PGA was influenced by the target and the distractor, consistent with an averaging strategy. When the target was known (KBYG), PGA reflected only the target. Overall, these results suggest that when confounds are controlled, the brain plans grasping movements under uncertainty by averaging possible actions.

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13 **Justyce Delaney Smith,** Hope Ouma, Mohammed Ali Ahmed, Sarah Afrouzi, and Richard E. Brown

Olfactory Reversal Learning and Long-Term Memory in Aged NRXN+/-, NRXN+/+ and DS5+/- Mice

Autism Spectrum Disorder (ASD) is characterised by genetic mutations which affect synaptic efficacy, resulting in reduced cognitive flexibility: difficulty in changing previously formed cognitions and behavioural patterns. Neurexin1 gene mutations have been associated with ASD, and we tested Neurexin1 mice in olfactory reversal learning, a test for cognitive flexibility. Male and female transgenic NRXN1+/- mice were compared with Wild-Type control mice (NRXN1 +/+), and two "recovery" strains that exclude splice site 5 (DS5- and DS5/DS5). During olfactory discrimination learning (A+/B-), DS5-mice showed higher error rates than other genotypes, but during reversal learning (B+/A-), there was no genotype difference. There were no significant sex differences in odour discrimination or reversal learning. The interaction between age and genotype showed that older Wild-Type and Transgenic mice were less likely to complete the study than younger mice and that old ds5/ds5 mice were more likely to complete the study than old mice of other genotypes. In memory tests done 7 and 30 days after reversal learning, mice preferred responding to the odour rewarded in reversal learning (B+), showing a recency effect. From these results, we conclude that age is an important factor in the development of deficits in cognitive function in NRXN mice.

14 **Callia Fieldhouse,** Kevin LeBlanc, and Heather Neyedli

Dissociating the Role of Visual and Reward Feedback in a Rapid Aiming Task

This study examined whether reward or visual endpoint feedback is required to guide strategic aiming toward a model-predicted optimum in a risky reaching task. Nine right-handed participants completed two experimental conditions (2 × 200 trials) that manipulated feedback type (Visual Only vs. Reward Only), with order counterbalanced. In the Visual Only condition, participants received full visual feedback of their hand and movement throughout the reach but were not shown the point outcome of each trial. In the Reward Only condition, vision was blocked during the movement, but participants were shown the point outcome after each reach. Optimal aim points were computed using a decision-theoretic model based on each participant's endpoint variability. Measures included mean horizontal endpoint, deviation from the model optimum, endpoint variability, reaction time, and movement time. Reward feedback led to aiming closer to the model optimum, whereas visual feedback produced larger deviations but reduced endpoint variability. Reward feedback was also associated with faster movement times, but no differences in reaction time. These findings suggest that reward signals promote value-directed aiming, whereas visual feedback enhances precision without reliably shifting aim toward the expected-value optimum, highlighting the distinct contributions of each feedback modality to motor decision-making.

15 **Audra Gezen,** Richard Drake, Catrina MacPhee, Joelle Safatli, Ariel Coish, Anne-Sophie Champod, and Gail Eskes

Measuring Spatial Neglect with Peg-the-Mole Performance

Spatial neglect (SN) is a common attentional disorder following right-hemisphere stroke, with reduced awareness of and/or responding to stimuli on the contralesional (left) side of space. SN has poor outcomes and thus effective treatments are critical. Prism adaptation (PA) is a promising treatment and we developed Peg-the-Mole (PTM) as a home-based, gamified, iPad tool for PA in which players reach and touch moles as they briefly appear on the screen. Although developed as a treatment, PTM's recording of reaction time (RT) and spatial error when touching targets makes it a potential outcome measure for monitoring neglect severity and treatment-related change. This study examined whether PTM performance is sensitive to neglect severity and its remediation across a 10-day PA intervention in ten participants with neglect wearing either 5-degree (active control) or 15-degree (therapeutic) prism goggles. Reaction time and error bias scores — reflecting differences in performance between left- and right-sided targets will be reported to examine these questions: 1) Does PTM performance reflect neglect severity on Day 1; 2) Does PTM performance improve over the 10 day treatment; 3) Does improvement depend upon strength of prism goggles? Positive findings could position PTM as a way to monitor spatial neglect during PA treatment.

16 **Skyla Junco** and Ian Weaver

Epigenetic Regulation of Transcription Factor SETBP1 by ATRX in SETBP1-HD

Autism spectrum disorder (ASD) is a neurodevelopmental condition, with approximately 20% of cases attributed to genetic mutations. Among implicated genes, SETBP1, a transcriptional modulator, and ATRX, a chromatin remodeling protein, play key roles in normal neurodevelopment. Emerging evidence suggests that ATRX may regulate SETBP1 expression, though the epigenetic mechanisms underlying this relationship remain unclear. This study examined whether ATRX influences Setbp1 expression through promoter DNA methylation and whether compensatory epigenetic responses occur in SETBP1 haploinsufficiency disorder (SETBP1-HD). Two experimental models were used: ATRX-overexpressing HEK293T cells and induced pluripotent stem cell (iPSC)-derived neuronal cultures carrying the pathogenic SETBP1 p.Glu545Ter variant, alongside matched controls. SETBP1 protein levels were measured by ELISA, mRNA expression by RT-qPCR, and promoter methylation by bisulfite pyrosequencing. ATRX overexpression significantly increased Setbp1 mRNA and reduced promoter methylation, supporting a role in promoting a transcriptionally permissive chromatin state. However, protein levels did not significantly increase, suggesting post-transcriptional buffering. In SETBP1-HD neurons, both mRNA and protein were reduced, while decreased promoter methylation indicated a compensatory response insufficient to restore expression. Overall, these findings suggest promoter methylation may regulate Setbp1 within biological limits, highlighting both the potential and constraints of epigenetic therapies in ASD.

17 **Sophie LeBlanc,** Maura Whitman, and Drew Weatherhead

Do Children Use Social Cues (e.g., Race and/or Sex/Gender) When Learning Phonological Variation?

Previous research shows that adults tend to prioritize race over another salient social cue (sex/gender) when learning a phonological variant (vowel shift). However, whether children use social cues to predict linguistic variation remains unclear. This study uses an artificial language task to investigate whether N=64 5-to-7-year-old children (preliminary data; target N=150) learned associations between phonological variation and speaker characteristics. Participants are randomly assigned to either a race or sex/gender condition, where the relevant cue predicted the correct response. On each of 32 trials, participants viewed a novel object between two images of closed doors and heard two variations for labelling the novel object (e.g., "Someone says [mɛlu], someone says [milu]"). Then, two speakers appeared before open doors, and participants were asked who said one of the variations, with immediate corrective feedback after every trial. After the experiment, participants were asked if anything helped them decide who said each word. A generalized estimating equation model revealed a significant main effect of condition, Wald $\chi^2 = 4.67$, $p = .031$, with higher accuracy in the sex/gender condition relative to race. These findings suggest that sex/gender may be a more salient cue at this developmental stage when learning phonological variation across different speakers.

18 **Kirsty Longino,** Shuqi Yu, and Hélène Deacon

Who Benefits from Digital Reading? Comparing Digital and Paper Reading Comprehension in Typical and Struggling Readers

This study examines digital reading comprehension across three reading profiles in Grade 6 students: typical readers, students with dyslexia, and poor comprehenders. We examined whether students' reading comprehension differs between digital and paper-based formats, how each format uniquely affects the three reading profile groups, and whether patterns of hyperlink use vary between these groups. Participants included 230 English-speaking children from Nova Scotia (M_{age} = 11 years, 7.5 months; SD = 3.46 months). Students with dyslexia and poor comprehenders were identified using standardized decoding and reading comprehension measures, with age- and nonverbal-matched typical readers as controls. All participants completed a paper-based reading comprehension test and a parallel digital version incorporating scrolling and hyperlinks that provided definitions to words. Results showed that across all Grade 6 participants, students demonstrated significantly higher comprehension when reading on paper than digitally. When comparing reading profiles, analyses showed that dyslexic and typical readers had lower reading comprehension digitally, whereas poor comprehenders performed better digitally, potentially benefiting from availability of hyperlinks. No significant differences were found in hyperlink use across groups. These findings suggest that digital reading enhances reading comprehension when features are helpful but may hinder it when features provide no meaningful support. This identified barriers and opportunities in digital reading with implications for future research and classroom practices.

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19 **Erin MacDougall,** Ramin Rostampour, Sarah DeGrace, Sherry Stewart

Cannabis Craving While Anticipating a Trauma Reminder: The Roles of Cannabis Coping Motives and Anxiety Sensitivity

Posttraumatic stress disorder (PTSD) is associated with increased cannabis use, often as a means of coping with trauma-related distress. Trauma-exposed individuals may experience heightened craving in anticipation of recalling a traumatic event. We investigate whether cannabis coping motives serve as a mediator between PTSD symptom severity and two aspects of negative anticipation elicited (NAE) cannabis craving: relief and reward, along with overall craving. We then examine whether anxiety sensitivity (AS) moderates this model. A sample of 202 trauma-exposed, cannabis-using adults completed validated measures of PTSD symptoms, cannabis coping motives, AS, and cannabis craving after being warned they might be asked to recall a distressing personal trauma. Structural equation modelling was used to test the hypotheses. Coping motives mediated the association between PTSD symptom severity and all components of NAE cannabis craving; the mediation model accounted for more variance in relief-oriented craving (43%) than in reward (10%). AS was positively correlated with PTSD symptom severity and coping motives but did not moderate the indirect effect on total cannabis craving. These findings suggest that coping-motivated craving may be particularly relevant to anticipatory, relief-oriented cannabis craving among trauma-exposed individuals. We suggest that AS alternatively serves as a transdiagnostic risk factor.

20 **Rhea Mehta** and Lori Wozney

Posttraumatic Stress Disorder Symptom Severity and Coping-Motivated Substance Use Among Canadian Youth with Interpersonal and Non-Interpersonal Trauma

Trauma type-interpersonal versus non-interpersonal-has been identified as an important predictor of posttraumatic stress disorder (PTSD) severity and may shape patterns of coping-motivated substance use during adolescence and emerging adulthood. The present study examined whether trauma type predicted PTSD symptom severity and coping-motivated alcohol and cannabis use among Canadian youth aged 15 to 25 with probable PTSD. A cross-sectional analysis of baseline data from a randomized controlled trial was conducted ($N = 42$). Index traumas were manually coded as interpersonal ($n = 36$) or non-interpersonal ($n = 6$). PTSD symptom severity was measured with the PTSD Checklist for DSM-5, and coping motives were assessed using the Brief Alcohol Motives Measure and Brief Cannabis Motives Measure coping subscales. Mann-Whitney U tests were used to examine group differences. Youth with interpersonal index traumas demonstrated significantly higher PTSD symptom severity than those with non-interpersonal traumas ($U = 173.00$, $p = .017$, $r = .36$). No significant group differences emerged for coping-motivated alcohol or cannabis use; the cannabis comparison trended contrary to prediction. Findings are discussed in relation to trauma-type frameworks and implications for scalable digital interventions for trauma-exposed youth.

21 **Ella Parsons,** Curve Lake Cultural Centre, Jacob van Haaften, and Shannon Johnson

Wellbeing and Connection with Creation in Curve Lake First Nation

For many Indigenous Peoples, wellbeing is inseparable from relationships with the natural world (Creation). Indigenous Knowledge systems have long emphasized the importance of connection with Creation for mental health. However, Indigenous-led, community-guided research remains limited. In collaboration with Curve Lake First Nation (CLFN), this study examined (1) whether connection with Creation differs by residential location, and (2) whether connection with Creation is associated with satisfaction with life, psychological distress, positive affect, and negative affect. Participants (N = 110), self-identified CLFN members aged 16–82, completed an online survey including the community-developed Curve Lake First Nation Connection with Creation Index (CLFN-CCI) and standardized wellbeing measures. A Kruskal-Wallis test indicated no significant differences in connection with Creation across residential location groups. Spearman's rank correlations revealed no significant associations between connection with Creation and wellbeing variables. High overall levels of connection with Creation and limited variability in scores may have constrained the detection of relationships, while unequal sizes residential location group sizes likely reduced statistical power. Despite non-significant results, this study offers important community-specific insights. Future research should prioritize larger, more balanced samples and the integration of Indigenous Knowledge systems when examining relationships between land-based connection and wellbeing.

22 **Momina Raja** and Antonina Omisade

Added Value of Etomidate Speech and Memory Test (eSAM) in Surgical Decision Making for Epilepsy

Surgical treatment of medically refractory mesial temporal lobe epilepsy (mTLE) provides patients with the best chance of seizure freedom but may result in post-operative anterograde amnesia (POAA). Neuropsychological testing (NP) can identify patients at high risk for POAA, but etomidate speech and memory (eSAM) test is currently the gold standard for predicting who will actually develop it. However, eSAM is invasive, which necessitates a better delineation of referral criteria for this procedure. This study examined eSAM results for patients with two atypical pre-surgical memory profiles, which identified them as at high risk for POAA to determine which profile(s) best coincided with eSAM findings versus when eSAM provided information over and above other clinical data. A retrospective chart review of eight patients (age=38.25 years; range=19-59 years; 62% females) with mTLE who underwent pre-surgical NP and eSAM procedure was completed. Memory scores from NP, clinical report interpretation, and eSAM findings were reviewed. Predicted memory change based on NP alone and based on NP with eSAM results were compared. There was no clear association between any specific high risk profiles and eSAM findings. This study supports continued use of eSAM to predict POAA in patients with all atypical memory profiles on NP.

23 **Aether Sangster,** Ray Klein, and Nick Murray

Spatial and Verbal Working Memory Contributions to Typing Performance

Typewriting is an important skill for communication and past research has revealed various psychological contributors to performance. Here we aim to understand the contributions of spatial and verbal working memory (WM) to speed and accuracy in this skill which consists of spatial and verbal elements. We compared typewriting performance under spatial and verbal WM loads to understand the role of WM in the process. Participants transcribed text under a time limit, in a control condition or while holding spatial or verbal items in WM. In Experiment 1 (N = 36), we compared speed and errors between these conditions and were surprised to find no significant performance differences from loading either subsystem. Experiment 2 (now in progress) modifies Experiment 1 to strengthen WM loading, investigate experience and automaticity as potential protectors against WM loading, and additionally characterize typewriting habits in 2026 by surveying task participants. So far, Experiment 2 supports our inference from Experiment 1, that typing is automatic enough to resist WM loading. Through these experiments, we are also compiling publicly available datasets of typewritten, transcribed text.

24 **Alexandra Seaman,** Isaiah Seale, and Richard Brown

Sex Differences in Olfactory Memory Strength and Transfer of Training in Neurexin1 Mouse Models of Autism Spectrum Disorder

Autism spectrum disorder (ASD) is characterized by impaired social skills and repetitive behaviour. Males are more likely to develop ASD, but data on sex differences in mouse models remains limited. This study investigated the effects of genotype, sex, and experience on memory strength and transfer of training in Nrxn1 mouse models. Male and female mice of four genotypes (Nrxn1+/+, Nrxn1+/-, DS5/-, DS5/DS5) underwent one-day Pavlovian odour discrimination training, followed by 24-hour, 7 day or 30 day memory tests. Time spent digging in odour pots associated with sugar reward (A+) or no reward (B-) difference scores (A-B), and preference scores (%A+) were assessed in a T maze after either memory test. Group N were trained in the T maze without prior Pavlovian conditioning. Males dug more in CS+ odour pots and had greater difference scores, however, no sex differences were found in CS- digging or preference scores, suggesting greater motivation in males rather than better memory. No genotype difference nor sex by genotype interactions were found, thus, Nrxn1+/- might not influence learning and memory. Despite no group differences in CS+ digging or sex by group interactions, greater scores for Group 30day relative to Group 7day and Group N across all other measures indicates that an additional reinforcement event improves odour discrimination and transfer following conditioning.

25 Evan Treffler, Thomas Snooks, Stephanie McAlister, Bill Le, Hayley Hamilton, Marvin Krank, Patricia Conrod, and Sherry Stewart

Personality Traits to Emotional and Behavioural Outcomes: Cognitive Mediation in Teens

Depression and conduct problems are prevalent adolescent mental health challenges linked to substantial impairment and long-term risk. Personality-based prevention models propose that hopelessness and impulsivity increase vulnerability to emotional and behavioral difficulties through cognitive mechanisms, including negatively biased automatic thoughts. This secondary analysis of baseline data from the Canadian Underage Substance Use Prevention (CUSP) trial investigated whether personal failure and hostile automatic thoughts mediated links between personality traits and mental health outcomes in adolescents. The sample included 2,834 secondary school students from 27 schools across Canada. Self-report measures assessed hopelessness, impulsivity, automatic thoughts, depressive symptoms, and conduct problems. Missing data (15.1% on average across variables) were imputed across 10 samples. Structural equation modeling examined pathways linking personality traits, cognitive mediators, and outcomes. Results supported the hypothesized mediational framework. The dominant pathways were hopelessness to depressive symptoms through failure thoughts, and impulsivity to conduct problems through hostile thoughts. Smaller cross-domain indirect effects emerged. Direct effects remained significant, indicating partial mediation. These findings support targeting cognitive processes in selective personality-based prevention programs like PreVenture.

26 Maura Whitman, Emma Reeves, Ali Dyack, Rebeka Workye, and Drew Weatherhead

Comparing Attitudes Toward Diverse and Homogeneous Groups Characterized by Race, Accent, and Sex/Gender Among Primary-School-Aged Children

Sex/gender, race, and accent have been shown to influence children's social preferences. However, this work primarily examines children's attitudes toward individuals, which are often used to infer their beliefs about groups. To better understand children's group-level social judgements, we investigated their attitudes toward homogeneous and diverse groups. 5-8-year-old children ($N = 68$) completed three blocks of trials varying by social dimension: sex/gender, race, and accent. On each trial, participants saw cartoon images of groups according to the social dimension being evaluated within each block (e.g., race): one group was homogeneous (matched to child), and the other was diverse (containing members varying on the relevant social dimension). Participants rated how much they liked each group using a 5-point Likert scale ranging from 1 (really don't like) to 5 (really like) with the visual aid of emojis. A linear mixed-effects model revealed significant main effects of group and social property. Across dimensions, homogeneous ingroups were rated higher than diverse groups, $F(1, 1562) = 38.48$, $p < .001$, while accent-based groups received the lowest ratings overall, $F(2, 1558) = 5.99$, $p = .003$. Follow-up research will include homogeneous outgroups to determine whether these findings reflect a preference for homogeneous ingroups or homogeneity.

27 **Deniz Yildirim,** Bilikis Banire, Dawson Sutherland, Youna McGowan, Sherry H. Stewart, Raymond M. Klein, and Sandra M. Meier

Comparison of Visual Tasks for Assessing Attentional Bias in Cannabis Users and Non-Users

Attentional bias toward substance-related cues is a key feature of addiction and may contribute to the continued use of substances, such as cannabis. However, the sensitivity of different paradigms for detecting this bias remains underexplored. This study compared three visual attentional bias paradigms: dot probe (DP), eye movement (EM), and MouseView.js (MV) in detecting cannabis-related attentional biases among cannabis users and non-users. Thirty-one participants (i.e., 16 users and 15 non-users) completed all three paradigms. The EM and MV tasks revealed significant stimulus effects, with non-users showing greater attention to neutral images, while cannabis users exhibited more evenly distributed attention across stimuli. In the EM task, a significant stimulus x severity interaction indicated reduced attentional differentiation with increased cannabis use severity. MV results followed a similar, but less precise pattern. In contrast, the DP task showed minimal group or interaction effects and lacked sensitivity to detect attentional differences. These findings suggest that EM-based measures are more sensitive for detecting cannabis-related attentional biases, particularly when accounting for cannabis use severity. MV showed some alignment with EM patterns but demonstrated weaker effects. The DP task, despite its widespread use, may be insufficient for capturing cannabis-related attentional differences.

28 **Gabrielle Zeller,** Ali Dyack, Amelia MacEachen, and Drew Weatherhead

First Impressions Are Lasting: How Children Use Clothing to Infer Group-Level Knowledge

Clothing is used to communicate identity, group membership, and even the knowledge one holds. Although adults understand that knowledge is retained even when appearances change (e.g., a chef without a uniform still knows how to cook), it is unclear whether children know to attribute shared knowledge to the individual, and not the clothing itself. Thus, the current study investigates where children are attributing clothing-based group level knowledge. In Experiment 1, 81 5–9-year-olds were introduced to novel objects that required either occupational or cultural knowledge. Object expertise was matched to a group of people with distinct clothing. Children were asked forced-choice questions about a specific group member's knowledge of the objects before and after changing clothes. To gain more nuanced participant responses, Likert scale and explanation questions were added to Experiment 2. In both experiments, children intuitively used clothing to infer expertise but tended to believe that knowledge was retained even after clothing changed. Further, when children initially assumed that the character lacked knowledge for an object, they were unlikely to infer that she would gain it later when wearing the associated clothing. Thus, children make assumptions based on clothes but are slow to update this knowledge when appearances change.